

ARTICLE TYPE

Sound Similarity in a Large Corpus of YM2612 FM Presets: From Register Space to Perceptual Distance

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Abstract

Frequency modulation (FM) synthesis on Yamaha FM chips underlies a large body of video-game and electronic music, yet systematic evidence on how register-level parameter changes relate to perceived timbre remains sparse. This paper studies sound similarity over a large corpus of real-world YM2612 presets extracted from Mega Drive / Genesis games. **Scope boundary:** the analysis starts from scratch; the only inheritance from prior repository work is the source instrument CSV. We render presets under a fixed protocol, extract audio descriptors and embedding spaces, and evaluate seven experiments (E1–E7) addressing five research questions (RQ1–RQ5). E1 preregisters a falsifiable hypothesis from Chowning’s FM theory: at equal parametric distance, modulator total-level (TL) changes should induce larger perceptual distance than carrier TL changes. Hardware validation notes the instrument’s 8 MHz master clock versus the console’s ≈ 7.67 MHz clock.

1. Introduction

Yamaha four-operator FM (as in the YM2612 / OPN2) maps a compact register set to a wide range of spectra via operator routing (algorithms), ratios, envelopes and total levels (Chowning, 1973). Game-music presets provide a culturally grounded, large-scale sample of how practitioners actually programmed the chip. Despite progress on synthesizer programming and sound matching Yee-King et al. (2018); Chen et al. (2022); Le Vaillant and Dutoit (2024), there is still limited work that (i) starts from a large *in-the-wild* chip-FM corpus, (ii) separates parameter-space geometry from audio-space geometry, and (iii) tests theory-driven predictions such as modulator versus carrier sensitivity.

Scope. This study does *not* reuse prior cluster labels, feature choices or hyperparameters from earlier exploratory work in the same repository. The sole inherited artifact is `data/processed/all_instruments_final.csv`. All representations, distances and evaluation protocols are defined and justified here.

Research questions.

- RQ1** At equal parametric distance, do modulator TL changes yield larger audio/perceptual distance than carrier TL changes, as predicted by Chowning (1973)?
- RQ2** Which effective YM2612 parameters most strongly control timbral change under controlled perturbations?

- RQ3** Do audio descriptor / embedding spaces recover known perceptual dimensions of timbre for FM (cf. McAdams et al. (1995))?
- RQ4** How do neighbourhoods in parameter space and in audio space relate (rank correlation; MFCC-based distances)?
- RQ5** Do automatic distances agree with human odd-one-out judgments?

Contributions (claimed). Corpus curation and rendering protocol; multi-condition similarity spaces; pre-registered E1 and evaluation suite E1–E7 with clustered inference. Morphing / PCA macro-controls are discussed only as future outlook (Section 8) and are *not* claimed as contributions.

2. Related Work

FM synthesis and programming. Chowning’s formulation (1973) remains the theoretical backbone. Classical optimization for FM tone matching includes genetic and related search (Horner et al., 1993; Dahlstedt, 2001, 2007). More recent interactive and evolutionary systems continue this line (Kreković, 2019).

Automatic synthesizer programming and sound matching. Deep and hybrid approaches estimate or search parameters for VSTs and FM instruments (Yee-King et al., 2018; Chen et al., 2022; Yang et al., 2023; Masuda and Saito, 2023; Roma, 2024). Latent preset models for DX7-class FM include SPINVAE / SPINVAE-2 (Le Vaillant et al., 2020, 2021; Le Vail-

lant and Dutoit, 2024). Differentiable FM and DDSP-style pipelines offer alternative routes (Caspe et al., 2022; Hayes et al., 2024). Neural proxies for black-box synths provide perceptually informed preset embeddings (Combes et al., 2025).

Timbre descriptors and corpora. The Timbre Toolbox (Peeters et al., 2011) and related MDS / perceptual studies (McAdams et al., 1995; Terasawa et al., 2005; Schubert and Wolfe, 2006) motivate descriptor choices. Chip and game-music datasets include NES-MDB (Donahue et al., 2018) and YM2413-MDB (Choi et al., 2022). Large synthetic preset resources such as synth1B1 (Turian et al., 2021) complement hand-authored banks. Visual exploration tools for audio collections include AudioStellar (Garber et al., 2021). Phase vocoder and related DSP tools support analysis pipelines Driedger et al. (2014); Cáceres (2007).

Differentiation from VibeFM. VibeFM (Rau et al., 2026) is a NIME interface for *sampling and visually exploring* FM parameter combinations, ordering glyphs by audio similarity to support serendipitous discovery during composition. Our work differs in goal and evidence standard: we analyse a large corpus of *existing game presets* on the YM2612, preregister theory-driven tests (E1), and report a multi-experiment evaluation (E1–E7) aimed at MIR claims about parameter–timbre geometry rather than an interactive composition tool. The two lines are complementary: VibeFM targets creative workflow; we target measurable structure in a culturally grounded chip-FM corpus.

3. Corpus

We start from the instrument CSV (all_instruments_final.csv) and rebuild a research corpus from scratch: validation/quarantine rules, configuration hashing, timbral-core grouping and TL-variant groups used by E1. Splits and seeds are fixed in configuration (research/configs/).

4. Method

4.1 Rendering

Presets are rendered through a YM2612 emulation core under a fixed note, velocity, duration and gain protocol (research/configs/render.yaml). Audio is stored as FLAC with a render manifest for provenance.

Hardware clock note. Canonical Mega Drive / Genesis YM2612 clocking is approximately 7.67 MHz (derived from the NTSC colour subcarrier chain). The hardware instrument used for validation runs at 8 MHz (a standard crystal). For identical registers, output frequency scales linearly with master clock, so an 8 MHz instrument plays $\approx 4.3\%$ sharp relative to 7.67 MHz ($\approx +73$ cents), with envelope times shortened by the same factor. We treat

this as an analytically invertible resampling identity in the digital domain and report residuals attributable to analogue / DAC behaviour separately (research/configs/hardware_validation.yaml).

4.2 Features

We extract hand-crafted descriptors (Timbre Toolbox-inspired and loudness-aware paths) and optional learned embeddings under versioned configs (research/configs/features.yaml).

4.3 Similarity spaces

We construct multiple conditions (parameter baseline after inertness masking; descriptor spaces; embedding spaces) with declared metrics and top- k probes. Full pairwise matrices are never materialised for $\sim 38k$ configurations; rank correlations use stratified blocked sampling (E4).

5. Evaluation

Clustered inference (within timbral core), Benjamini–Hochberg FDR and bootstrap effect-size CIs are declared in research/configs/eval.yaml. No numerical results are reported in this scaffold.

5.1 E1 – Modulator vs. carrier TL (Chowning)

Preregistered hypothesis: at equal parametric distance, pairs differing only in modulator TL show significantly larger perceptual/audio distance than pairs differing only in carrier TL, because modulator TL controls modulation index (spectral bandwidth) whereas carrier TL is essentially amplitude Chowning (1973). Primary loudness path and null-result protocol are fixed before analysis.

5.2 E2 – Perturbation sensitivity

Budgeted one-parameter perturbations of effective registers; sensitivity curves per parameter family.

5.3 E3 – McAdams-style dimensions

Canonical correlation / alignment of principal audio-space axes with attack time, spectral centroid and spectral flux (McAdams et al., 1995; Peeters et al., 2011).

5.4 E4 – Rank correlation across spaces

Spearman correlation between parameter-space and audio-space distances on stratified pair samples (no full matrix).

5.5 E5 – MFCCD anchors

MFCC distance distributions with published Sound2Synth-style thresholds as external anchors (Chen et al., 2022).

5.6 E6 – Listening test

Triadic odd-one-out; if infeasible, RQ5 is removed rather than left unanswered.

5.7 E7 — Game footprint (secondary)

Same-game retrieval metrics, aggregated per game to avoid size bias; declared confounded (OSTs intentionally mix timbres).

6. Results

TODO: insert from artifacts/results (E1–E7 tables and figures). Placeholder figure slots:

- Figure ?? — E1 modulator vs. carrier TL separation
- Figure ?? — E2 perturbation sensitivity
- Figure ?? — space neighbourhood / embedding overview

7. Discussion and Limitations

Limitations include emulator–hardware residual error after clock alignment; descriptor vs. embedding disagreements; confounding in E7; and dependence of RQ5 on the listening study.

8. Future Outlook

PCA-based morph / macro-controls over effective parameters are a natural *downstream* application of a validated similarity space (Capacity A in the research package). They are **not** claimed as contributions of this paper; any morph results belong to follow-up systems work, not to the MIR evaluation here.

9. Conclusion

We outlined a from-scratch study of sound similarity for YM2612 game presets, centred on RQ1–RQ5 and experiments E1–E7, with a Chowning-grounded E1 hypothesis and an explicit 8 MHz vs. 7.67 MHz hardware-clock note.

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